Jan Škvrna

Weakly Supervised 3D Perception & Generative 3D Reconstruction

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jskvrna.github.io | GitHub | Google Scholar | LinkedIn

Available for research internships in 2027 with flexible timing and location.

I am a CTU Prague PhD student with four first-author papers at ECCV, ICCV, IJCV and WACV. My teams won two S23DR challenges at the CVPR Urban Scene Modeling Workshop. My research covers 3D object detection without human annotations and generative 3D reconstruction, and I currently work on 4D semantic occupancy.

Education & Research

PhD in Informatics Feb 2024 – Expected Feb 2028

CTU Prague, Visual Recognition Group (led by Prof. Jiří Matas), advisor Lukáš Neumann, Ph.D.

MSc in Cybernetics and Robotics, with Honors 2021 – 2024

CTU Prague, Dean's Award for Master's Thesis

Summer schools: EEML (2026), ICVSS (2025).

Publications

WireDiT: A Vertex-Slot Diffusion Transformer for 3D Wireframe Reconstruction

***J. Škvrna**, M. Purkrábek, L. Neumann.* **WACV 2027, accepted**.

- Flow-matching transformer with 64 vertex slots conditioned on point-anchored scene tokens, predicting 3D building wireframes (vertices and edges) from sparse street-level SfM points, depth and semantics.
- **State of the art on HoHo22k** (private test set vertex F1 0.791 vs. 0.751, HSS 0.654 vs. 0.648 for the next-best method). On Building3D Tallinn, **23% lower average corner offset** (0.17 vs. 0.22 m, EdgeDiff+GREO).

MonoSOWA++: Universal Autolabeling Pipeline for Monocular 3D Object Detection

***J. Škvrna**, L. Neumann.* **IJCV 2026**. Paper | Code

- Journal extension of MonoSOWA with pedestrians, cyclists and pseudo-label confidence scoring. For cars, autolabel **3D AP 57.6 vs. 47.1** (MonoSOWA) and 35.3 (VSRD) on KITTI-360 (Easy, IoU 0.5).
- Pedestrian 3D AP 23.6 with no human labels vs. 28.1 for fully supervised MonoDETR (KITTI, Easy, IoU 0.3).

MonoSOWA: Scalable Monocular 3D Object Detector Without Human Annotations

***J. Škvrna**, L. Neumann.* **ICCV 2025**. Paper | Code

- Autolabeling for monocular 3D detection without human annotations that separates object from ego motion across a video and normalizes focal length across cameras, at 1.3 s per frame vs. about 15 min for VSRD.
- MonoDETR pretrained on its KITTI-360 pseudo-labels and fine-tuned on 25% of KITTI labels reaches **25.1 3D AP vs. 20.6** when trained from scratch on all labels (Car, moderate, IoU 0.7).

TCC-Det: Temporarily Consistent Cues for Weakly-Supervised 3D Detection

***J. Škvrna**, L. Neumann.* **ECCV 2024**. Paper | Code

- LiDAR 3D detector trained without human 3D labels via temporal consistency and robust template fitting.
- KITTI **3D AP 75.3 vs. 13.3** for the best prior label-free method, which is **89% of fully supervised** Voxel-RCNN at 84.5 (Car, moderate, IoU 0.7). Waymo Open 3D AP 70.0 vs. 94.2 supervised (Level 1, IoU 0.5).

Awards

1st place of 27 teams, S23DR 3D Reconstruction Challenge (US$5,000 prize) 2026

CVPR Urban Scene Modeling Workshop, WireDiT solution, team of three. Report | Project

1st place, S23DR 3D Reconstruction Challenge (US$10,000 prize) 2025

CVPR Urban Scene Modeling Workshop, two-stage PointNet pipeline, team of two. Report | Project

Experience, Service & Talks

Junior Embedded Software Developer, Poll s.r.o. 2020 – 2023

Real-time C/C++ ARM firmware for HVAC and high-voltage DC/DC converters deployed on trains.

Lab Instructor, Deep Learning Essentials, CTU Prague 2025 – Present

Reviewer: ICCV 2025, BMVC 2026, AAAI 2027. **Talks:** CMP Colloquium (2026), CVWW (2025).

Skills

Research: Diffusion and flow matching, multi-view geometry, 3D reconstruction and detection, 4D occupancy.

Engineering: Python, PyTorch, multi-GPU training, Slurm, C/C++, Open3D, PyTorch3D, OpenCV.